THIRUPATI GOPI NAGA VIGNESH

Aerospace & Computational Software Engineer

gopinagavignesh375@gmail.com
+91 7013378520
Hyderabad, India
linkedin.com/in/vignesh375
github.com/perfectalone375

PROFILE

Computational Aerospace Software Engineer and AI Evaluation Specialist with over 2 years of professional experience in scientific programming, trajectory dynamics, and AI code generation benchmarking. Skilled in Python, NumPy, SciPy, Pytest, Docker, and Git/GitHub CI/CD workflows. Experienced in developing high-fidelity mathematical models for flight dynamics, atmospheric thermodynamics, and numerical ODE solvers. Proven track record authoring SWE-bench style evaluation tasks, validating frontier AI coding models, and maintaining open-source engineering packages under MIT licenses.

PROFESSIONAL EXPERIENCE

Freelance (Self-Employed), Software Engineer & AI Evaluation Specialist
01/2026 – Present
Hyderabad, India

- Designed and authored high-fidelity coding benchmark tasks to train and evaluate frontier AI models following SWE-bench evaluation methodology.
- Created deterministic unit tests (fail-to-pass / pass-to-pass), realistic prompt problem descriptions, and minimal golden patches across scientific Python codebases.
- Conducted AI model evaluation, code generation assessment, data annotation, and RLHF validation for technical engineering workloads.
- Containerized test harnesses using Docker and automated CI execution to guarantee reproducible, sandboxed evaluation.
- Founded and built Froscel, an AI-powered talent platform, leading architecture design, MVP development, and full-stack software delivery.

AZISTA COMPOSITES PRIVATE LIMITED, Design & Computational Engineer
05/2024 – 01/2026
Hyderabad, India

- Developed Python and Excel-based automation scripts for engineering workflows, reducing manual analysis time and eliminating numerical calculation errors.
- Engineered structural validation, load-path simulation, and performance analysis algorithms for critical aerospace programs including BrahMos Battery Frames, Solar Panel Assemblies, and Navy Doors.
- Conducted material testing data analysis using UTM instrumentation, developing automated data extraction and reduction scripts in Python.
- Collaborated with cross-functional engineering teams to validate physical and digital twin models against strict defense manufacturing standards.
- Prepared detailed parametric models and technical documentation ensuring full compliance with ASME and aerospace industry specifications.

Directorate of Special Projects, RCI Campus, DRDO, UAS & Satellite Systems Intern
02/2024 – 03/2024
Hyderabad, India

- Contributed to the design, telemetry data logging, and subsystem integration of Quadsat UAV systems, supporting propulsion and sensor communication.
- Assisted in satellite communication studies, orbital trajectory calculations, and signal analysis for Low Earth Orbit (LEO) satellite tracking.
- Conducted system verification testing, data analysis, and technical documentation for remote sensing antenna arrays.

EDUCATION

Bachelor of Technology, Aeronautical Engineering & Computational Modeling
12/2020 – 03/2024
MLR Institute of Technology | Percentage: **78%**

Intermediate Education (MPC), Master Junior College
06/2018 – 04/2020
Mathematics, Physics, Chemistry | Percentage: **95%**

Secondary Education, SaiGiri School
06/2016 – 04/2018
CGPA: **9.2**

SKILLS

Programming & Software: Python (NumPy, SciPy, Matplotlib), C++, SQL, Bash scripting, Git, GitHub Actions (CI/CD), Docker, Docker Compose, Pytest, Ruff linter, Test-Driven Development (TDD), Object-Oriented Programming (OOP)

AI & Benchmark Engineering: SWE-bench Task Authoring, LLM Code Generation Evaluation, Fail-to-Pass Test Harnesses, Regression Testing, Prompt Engineering, RLHF Annotation, Model Quality Assessment

Computational & Aerospace: 3-DoF / 6-DoF Trajectory Dynamics, 1976 U.S. Standard Atmosphere Modeling, Compressible Aerodynamics, Transonic Wave Drag Divergence, Runge-Kutta 4th Order (RK4) Numerical Solvers, Telemetry Processing

CAD & Engineering Tools: Siemens NX CAD | CATIA | SolidWorks | Fusion 360 | Autodesk Inventor | ANSYS | 3D Modeling | GD&T | ASME Y14.5 | Tolerance Analysis

Professional: Cross-Functional Engineering Collaboration | Code Review & Git PR Workflows | Root Cause Analysis | Technical Documentation | High-Reliability Software Systems

KEY PROJECTS

aero-flight-dynamics (Open Source — github.com/perfectalone375/aero-flight-dynamics)
Architected an open-source 3-DoF flight dynamics and trajectory simulation library in Python under the MIT license. Implemented the exact 7-layer 1976 U.S. Standard Atmosphere model (0–86 km) with Sutherland's dynamic viscosity law, compressible aerodynamic drag/lift, solid rocket propulsion mass depletion, and RK4 numerical solver. Built an automated 22-test suite in Pytest, multi-version GitHub Actions CI matrix, and containerized Docker execution. Authored SWE-bench AI benchmark tasks with fail-to-pass regression verification.

Multi-Purpose Autonomous UAV for Surveillance and Target Tracking
Engineered an intelligent UAV platform incorporating real-time computer vision object detection and autonomous target tracking algorithms to support advanced surveillance and monitoring operations.

Multi-Disciplinary UAV and Launch Surveillance Gliders
Engineered a UAV-based surveillance platform with deployable gliders, developing flight mechanics models and telemetry analysis tools to optimize aerial coverage, situational awareness, and range.

Design and Fabrication of Fixed Wing UAV
Designed, analyzed, and manufactured an autonomous fixed-wing UAV, applying aerodynamic lift-to-drag optimization, computational flight stability analysis, and lightweight structural principles.

BrahMos Battery Frame Structural Design & Numerical Verification
Engineered a lightweight composite battery frame assembly for defense missile applications, performing stress analysis, structural load simulations, and mechanical validation for metallic-composite interfaces.

Solar Panel Substrate for Aerospace Applications
Engineered lightweight composite-metal sandwich panels, conducting structural rigidity and vibration simulations to optimize strength-to-weight ratio and long-term durability in aerospace environments.

CERTIFICATES

- **Aircraft Design:** Certified by IIT Kanpur, NPTEL (01/2023 - 04/2023)
- **Introduction to Airplane Performance:** Certified by IIT Kanpur, NPTEL (07/2023 - 09/2023)
- **Flight Fury:** Certified by Cognizance IIT Roorkee (03/2023)
- **Drone Development Challenge:** Certified by SAE India (07/2023)
- **Aeromodeling Event:** Certified by Indian Institute of Aerospace Technology and Management (08/2017)

AWARDS & ORGANISATIONS

- **Essay Writing Competition:** 3rd Position organized by National Institute for Micro Small and Medium Enterprises.
- **Society of Automotive Engineers (SAE) INDIA:** Active Member (07/2023 – 07/2024).

LANGUAGES & INTERESTS

Languages: English, Telugu, Hindi | **Interests:** Computational Simulation, Space Science, Computer Science, AI Benchmarking, Cricket